



3 Best Approximations

In this chapter, we analyze fundamental questions of approximation. To this end, let $\mathcal{F}$ be a linear space, equipped with a norm $\|\cdot\|$. Moreover, $\mathcal{S} \subset \mathcal{F}$ be a non-empty subset of $\mathcal{F}$. To approximate one $f \in \mathcal{F} \setminus \mathcal{S}$ by elements from $\mathcal{S}$ we are interested in finding a $s^* \in \mathcal{S}$, whose distance to f is *minimal* among all elements from $\mathcal{S}$. This leads us to the definition of *best approximations*.

Definition 3.1. Let $\mathcal{F}$ be a linear space with norm $\|\cdot\|$. Moreover, let $\mathcal{S} \subset \mathcal{F}$ be a non-empty subset of $\mathcal{F}$. For $f \in \mathcal{F}$, an element $s^* \in \mathcal{S}$ is said to be a **best approximation** to f from $\mathcal{S}$ with respect to $(\mathcal{F}, \|\cdot\|)$, or in short: s^* is a **best approximation** to f , if

$$\|s^* - f\| = \inf_{s \in \mathcal{S}} \|s - f\|.$$

Moreover,

$$\eta \equiv \eta(f, \mathcal{S}) = \inf_{s \in \mathcal{S}} \|s - f\|$$

is called the **minimal distance** between f and $\mathcal{S}$. ○

In the following investigations, we will first address questions concerning the existence and uniqueness of best approximations. To this end, we develop sufficient conditions for the linear space $\mathcal{F}$ and the subset $\mathcal{S} \subset \mathcal{F}$, under which we can guarantee for any $f \in \mathcal{F}$ the existence of a best approximation $s^* \in \mathcal{S}$ for f . To guarantee the uniqueness of s^* , we require *strict convexity* for the norm $\|\cdot\|$.

In the following discussion, we develop suitable sufficient and necessary conditions to characterize best approximations. To this end, we first derive *dual* characterizations for best approximations, giving conditions for the elements from the *topological dual space* $\mathcal{F}'$ of linear and continuous functionals on $\mathcal{F}$.

This is followed by *direct* characterizations of best approximations, where we use directional derivatives (*Gâteaux derivatives*) of the norm $\|\cdot\|$. On that occasion, we consider computing directional derivatives of relevant norms explicitly.

To study the material of this chapter (and for the following chapters) we require knowledge of elementary results from optimization and functional analysis. Therefore, we decided to explain a selection of relevant results. But for further reading, we refer to the textbook [33].

Before we address theoretical questions concerning the existence and uniqueness of best approximations, we first discuss one elementary example, which will illustrate relevant scenarios and phenomena.

Example 3.2. For $\mathcal{F} = \mathbb{R}^2$, let $\mathcal{S} = \{x = (x_1, x_2) \mid 2 \leq \|x\|_2 < 3\} \subset \mathbb{R}^2$ be a concentric circle around the origin. Moreover, let $f_\alpha = (\alpha, 0) \in \mathbb{R}^2$, for $\alpha \in \mathbb{R}$. Now we wish to *best-approximate* f_α (according to Definition 3.1) by elements from $\mathcal{S}$. To do so, we first need to fix a norm on $\mathbb{R}^2$. To this end, we work with three different norms on $\mathbb{R}^2$:

- the 1-norm $\|\cdot\|_1$, defined as $\|x\|_1 = |x_1| + |x_2|$ for $x = (x_1, x_2)$;
- the Euclidean norm $\|\cdot\|_2$, defined as $\|x\|_2^2 = |x_1|^2 + |x_2|^2$;
- the maximum norm $\|\cdot\|_\infty$, defined as $\|x\|_\infty = \max(|x_1|, |x_2|)$.

We let $\mathcal{S}_p^* \equiv \mathcal{S}_p^*(f_\alpha)$ denote the set of best approximations to f_α with respect to $\|\cdot\| = \|\cdot\|_p$ at minimal distances $\eta_p \equiv \eta_p(f_\alpha, \mathcal{S})$, for $p = 1, 2, \infty$. For the construction and characterization of best approximations to f_α we distinguish different cases (see Fig. 3.1).

Case (a): Suppose $\alpha \geq 3$. In this case, we have

$$\eta_p = \inf_{s \in \mathcal{S}} \|s - f_\alpha\|_p = \alpha - 3 \quad \text{for } p = 1, 2, \infty,$$

where

$$\|s - f_\alpha\|_p > \inf_{s \in \mathcal{S}} \|s - f_\alpha\|_p = \alpha - 3 \quad \text{for all } s \in \mathcal{S},$$

i.e., there is *no* best approximation to f_α from $\mathcal{S}$, and so $\mathcal{S}_p^* = \emptyset$.

Case (b): Suppose $\alpha \in (0, 2)$. In this case, we have

$$\eta_1 = \eta_2 = 2 - \alpha \quad \text{and} \quad \eta_\infty = \frac{\sqrt{8 - \alpha^2} - \alpha}{2}$$

and, moreover, $\mathcal{S}_p^* = \{(2, 0)\}$ for $p = 1, 2$ and $\mathcal{S}_\infty^* = \left\{ \left(\frac{\sqrt{8 - \alpha^2} + \alpha}{2}, \pm \frac{\sqrt{8 - \alpha^2} - \alpha}{2} \right) \right\}$.

Case (c): Suppose $\alpha = 0$. In this case, we have

$$\eta_1 = \eta_2 = 2 \quad \text{and} \quad \eta_\infty = \sqrt{2}$$

where $\mathcal{S}_1^* = \{(\pm 2, 0), (0, \pm 2)\}$, $\mathcal{S}_2^* = \{x \in \mathcal{S} \mid \|x\|_2 = 2\}$, $\mathcal{S}_\infty^* = \{(\pm\sqrt{2}, \pm\sqrt{2})\}$. Therefore, there exists, for any of the three norms $\|\cdot\|_p$, $p = 1, 2, \infty$, a best approximation to f_0 . In either case, however, the best approximations are not unique. For $\|\cdot\|_2$ there are even uncountably many best approximations to f_0 .

Case (d): For $\alpha \in [2, 3)$ we have $f_\alpha \in \mathcal{S}$ and so $\mathcal{S}_p^* = \{f_\alpha\}$ with $\eta_p = 0$.

For any other case, i.e., for $\alpha < 0$, we can analyze the set of best approximations by using one of the (symmetric) cases (a)-(d). $\diamond$

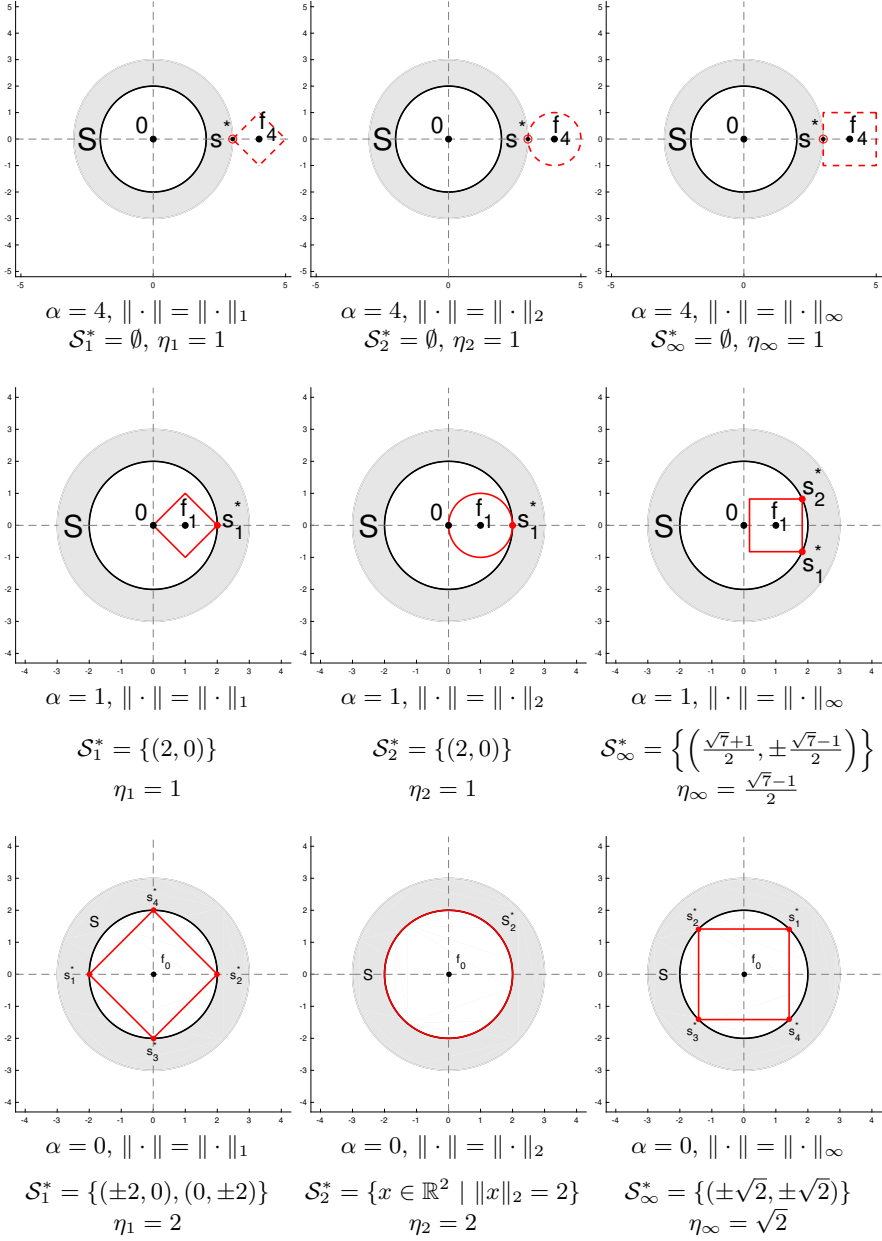


Fig. 3.1. Approximation of $f_\alpha = (\alpha, 0) \in \mathbb{R}^2$, for $\alpha = 4, 1, 0$, by elements from the approximation set $S = \{x = (x_1, x_2) \mid 2 \leq \|x\|_2 < 3\} \subset \mathbb{R}^2$ and with respect to the norms $\|\cdot\|_p$ for $p = 1, 2, \infty$ (see Example 3.2).

3.1 Existence

In the following discussion, the notions of *compactness*, *completeness*, and *continuity* play an important role. We assume that their definitions and further properties are familiar from analysis. Nevertheless, let us recall the continuity of functionals. Throughout this chapter, $\mathcal{F}$ denotes a linear space with norm $\|\cdot\|$.

Definition 3.3. A functional $\varphi : \mathcal{F} \rightarrow \mathbb{R}$ is said to be **continuous** at $u \in \mathcal{F}$, if for any convergent sequence $(u_n)_{n \in \mathbb{N}} \subset \mathcal{F}$ with limit $u \in \mathcal{F}$, i.e.,

$$\|u_n - u\| \rightarrow 0 \quad \text{for } n \rightarrow \infty,$$

we have

$$\varphi(u_n) \rightarrow \varphi(u) \quad \text{for } n \rightarrow \infty.$$

Moreover, φ is called **continuous on $\mathcal{F}$** , if φ is continuous at every $u \in \mathcal{F}$. ○

Now recall that any continuous functional attains its minimum (and its maximum) on compact sets. Any compact set is closed and bounded. The converse, however, is only true in *finite-dimensional* spaces.

For the discussion in this section, we need the continuity of norms. This requirement is already covered by the following result.

Theorem 3.4. Every norm is continuous.

Proof. Let $\mathcal{F}$ be a linear space with norm $\|\cdot\|$. Moreover let $v \in \mathcal{F}$ and $(v_n)_{n \in \mathbb{N}} \subset \mathcal{F}$ be a convergent sequence in $\mathcal{F}$ with limit v , i.e.,

$$\|v_n - v\| \rightarrow 0 \quad \text{for } n \rightarrow \infty.$$

Now, by the triangle inequality for the norm $\|\cdot\|$, this implies

$$|\|v_n\| - \|v\|| \leq \|v_n - v\| \rightarrow 0 \quad \text{for } n \rightarrow \infty$$

and therefore

$$\|v_n\| \rightarrow \|v\| \quad \text{for } n \rightarrow \infty,$$

i.e., $\|\cdot\|$ is continuous at $v \in \mathcal{F}$. Since we did not pose any further conditions on $v \in \mathcal{F}$, the norm $\|\cdot\|$ is continuous on $\mathcal{F}$. ■

The above result allows us to prove a first elementary result concerning the existence of best approximations.

Theorem 3.5. Let $\mathcal{S} \subset \mathcal{F}$ be compact. Then there exists for any $f \in \mathcal{F}$ a best approximation $s^* \in \mathcal{S}$ to f .

Proof. For $f \in \mathcal{F}$ the functional $\varphi : \mathcal{F} \rightarrow [0, \infty)$, defined as

$$\varphi(v) = \|v - f\| \quad \text{for } v \in \mathcal{F},$$

is continuous on $\mathcal{F}$. In particular, φ attains its minimum on the compact set $\mathcal{S}$, i.e., there is one $s^* \in \mathcal{S}$ satisfying

$$\varphi(s^*) = \|s^* - f\| \leq \|s - f\| = \varphi(s) \quad \text{for all } s \in \mathcal{S}.$$

■

From this result, we can further conclude as follows.

Corollary 3.6. *Let $\mathcal{F}$ be finite-dimensional, and $\mathcal{S} \subset \mathcal{F}$ be closed in $\mathcal{F}$. Then there exists for any $f \in \mathcal{F}$ a best approximation $s^* \in \mathcal{S}$ to f .*

Proof. For $s_0 \in \mathcal{S}$ and $f \in \mathcal{F}$ the non-empty set

$$\mathcal{S}_0 = \mathcal{S} \cap \{v \in \mathcal{F} \mid \|v - f\| \leq \|s_0 - f\|\} \subset \mathcal{S}$$

is closed and bounded, i.e., $\mathcal{S}_0 \subset \mathcal{F}$ is compact. By Theorem 3.5 there is one best approximation $s^* \in \mathcal{S}_0$ to f from $\mathcal{S}_0$, so that

$$\|s^* - f\| \leq \|s - f\| \quad \text{for all } s \in \mathcal{S}_0.$$

Moreover, for any $s \in \mathcal{S} \setminus \mathcal{S}_0$ we have the inequality

$$\|s - f\| > \|s_0 - f\| \geq \|s^* - f\|,$$

so that altogether,

$$\|s^* - f\| \leq \|s - f\| \quad \text{for all } s \in \mathcal{S},$$

i.e., $s^* \in \mathcal{S}_0 \subset \mathcal{S}$ is a best approximation to f from $\mathcal{S}$. ■

Corollary 3.7. *Let $\mathcal{S} \subset \mathcal{F}$ be a closed subset of $\mathcal{F}$. If $\mathcal{S}$ is contained in a finite-dimensional linear subspace $\mathcal{R} \subset \mathcal{F}$ of $\mathcal{F}$, i.e., $\mathcal{S} \subset \mathcal{R}$, then there exists for any $f \in \mathcal{F}$ a best approximation $s^* \in \mathcal{S}$ to f .*

Proof. Regard, for $f \in \mathcal{F}$, the finite-dimensional linear space

$$\mathcal{R}_f = \text{span}\{f, r_1, \dots, r_n\} \subset \mathcal{F},$$

where $\{r_1, \dots, r_n\}$ be a basis of $\mathcal{R}$. Then there exists, by Corollary 3.6 a best approximation $s^* \in \mathcal{S}$ to $f \in \mathcal{R}_f$, where in particular,

$$\|s^* - f\| \leq \|s - f\| \quad \text{for all } s \in \mathcal{S}.$$

■

The result of Corollary 3.7 holds in particular for the case $\mathcal{R} = \mathcal{S}$.

Corollary 3.8. *Let $\mathcal{S} \subset \mathcal{F}$ be a finite-dimensional subspace of $\mathcal{F}$. Then there exists for any $f \in \mathcal{F}$ a best approximation $s^* \in \mathcal{S}$ to f . ■*

In the above results concerning the existence of best approximations, we require that $\mathcal{S} \subset \mathcal{F}$ is contained in a finite-dimensional linear space. For the approximation in Euclidean spaces $\mathcal{F}$, we can refrain from using this restriction. To this end, the following geometric identities are of fundamental importance.

Theorem 3.9. *Let $\mathcal{F}$ be a Euclidean space with inner product $(\cdot, \cdot)$ and norm $\|\cdot\| = (\cdot, \cdot)^{1/2}$. Then the **parallelogram identity***

$$\|v + w\|^2 + \|v - w\|^2 = 2\|v\|^2 + 2\|w\|^2 \quad \text{for all } v, w \in \mathcal{F} \quad (3.1)$$

*holds. If $\mathcal{F}$ is a Euclidean space over the real numbers $\mathbb{R}$, then the **polarization identity***

$$(v, w) = \frac{1}{4} (\|v + w\|^2 - \|v - w\|^2) \quad \text{for all } v, w \in \mathcal{F}. \quad (3.2)$$

*holds. If $\mathcal{F}$ is a Euclidean space over the complex numbers $\mathbb{C}$, then the **polarization identity** holds as*

$$(v, w) = \frac{1}{4} (\|v + w\|^2 - \|v - w\|^2 + i\|v + iw\|^2 - i\|v - iw\|^2) \quad (3.3)$$

for all $v, w \in \mathcal{F}$.

Proof. Equations (3.1), (3.2) follow directly from the identities

$$\|v \pm w\|^2 = (v \pm w, v \pm w) = (v, v) \pm 2(v, w) + (w, w) = \|v\|^2 \pm 2(v, w) + \|w\|^2.$$

Likewise, the polarization identity (3.3) can be verified by elementary calculations. ■

For the geometric interpretation of the parallelogram identity, we make the following remark.

For any parallelogram the sum of square lengths of the four edges coincides with the sum of the square lengths of the two diagonals (see Fig. 3.2).

For the statement in Theorem 3.9, the converse is true, according to the theorem of Jordan¹ and von Neumann² [40].

Theorem 3.10. (Jordan-von Neumann theorem, 1935).

Let $\mathcal{F}$ be a linear space with norm $\|\cdot\|$, for which the parallelogram identity (3.1) holds. Then there is an inner product $(\cdot, \cdot)$ on $\mathcal{F}$, so that

$$(v, v) = \|v\|^2 \quad \text{for all } v \in \mathcal{F}, \quad (3.4)$$

i.e., $\mathcal{F}$ is a Euclidean space.

¹ PASCUAL JORDAN (1902-1980), German mathematician and physicist

² JOHN VON NEUMANN (1903-1957), Hungarian-US American mathematician

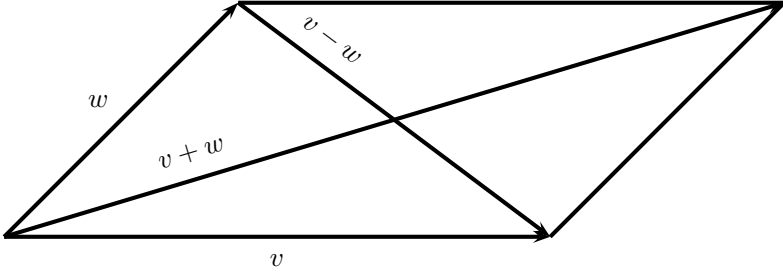


Fig. 3.2. On the geometry of the parallelogram identity (see Theorem 3.9).

Proof. Let $\mathcal{F}$ be a linear space over $\mathbb{R}$. By using the norm $\|\cdot\|$ of $\mathcal{F}$ we define a mapping $(\cdot, \cdot) : \mathcal{F} \times \mathcal{F} \longrightarrow \mathbb{R}$ through the polarization identity (3.2), i.e., we let

$$(v, w) := \frac{1}{4} (\|v + w\|^2 - \|v - w\|^2) \quad \text{for } v, w \in \mathcal{F}.$$

Obviously, we have (3.4) and so $(\cdot, \cdot)$ is positive definite. Moreover, $(\cdot, \cdot)$ is obviously symmetric, so that $(v, w) = (w, v)$ for all $v, w \in \mathcal{F}$.

It remains to verify the linearity

$$(\alpha u + \beta v, w) = \alpha(u, w) + \beta(v, w) \quad \text{for all } \alpha, \beta \in \mathbb{R}, u, v, w \in \mathcal{F}. \quad (3.5)$$

To this end, we note the property

$$(-v, w) = -(v, w) \quad \text{for all } v, w \in \mathcal{F}, \quad (3.6)$$

which immediately follows from the definition of $(\cdot, \cdot)$. In particular, we have

$$(0, w) = 0 \quad \text{for all } w \in \mathcal{F}.$$

Moreover, by the parallelogram identity (3.1) we obtain

$$\begin{aligned} (u, w) + (v, w) &= \frac{1}{4} (\|u + w\|^2 - \|u - w\|^2 + \|v + w\|^2 - \|v - w\|^2) \\ &= \frac{1}{2} \left(\left\| \frac{1}{2}(u + v) + w \right\|^2 - \left\| \frac{1}{2}(u + v) - w \right\|^2 \right) \\ &= 2 \left(\frac{1}{2}(u + v), w \right), \end{aligned}$$

which, for $v = 0$, implies

$$(u, w) = 2 \left(\frac{1}{2} u, w \right) \quad \text{for all } u, w \in \mathcal{F} \quad (3.7)$$

and thereby the *additivity*

$$(u, w) + (v, w) = (u + v, w) \quad \text{for all } u, v, w \in \mathcal{F}. \quad (3.8)$$

From (3.7), (3.8) we obtain for $m, n \in \mathbb{N}$ the identities

$$\begin{aligned} m(u, w) &= (mu, w) & \text{for all } u, w \in \mathcal{F} \\ \frac{1}{2^n}(u, w) &= \left(\frac{1}{2^n} u, w \right) & \text{for all } u, w \in \mathcal{F} \end{aligned}$$

by induction on $m \in \mathbb{N}$ and by induction on $n \in \mathbb{N}$, respectively.

In combination with (3.6) and (3.8) this implies the *homogeneity*

$$(\alpha u, w) = \alpha(u, w) \quad \text{for all } u, w \in \mathcal{F} \quad (3.9)$$

for all *dyadic* numbers $\alpha \in \mathbb{Q}$ of the form

$$\alpha = m + \sum_{k=1}^n \frac{\alpha_k}{2^k} \quad \text{for } m \in \mathbb{Z}, n \in \mathbb{N}, \alpha_k \in \{0, 1\}, 1 \leq k \leq n.$$

Since any real number $\alpha \in \mathbb{R}$ can be approximated arbitrarily well by a dyadic number, the continuity of the norm $\|\cdot\|$ implies the homogeneity (3.9) even for all $\alpha \in \mathbb{R}$. Together with the additivity (3.8) this implies the linearity (3.5). Therefore, $(\cdot, \cdot)$ is an inner product over $\mathbb{R}$.

If $\mathcal{F}$ is a linear space over $\mathbb{C}$, then we define $(\cdot, \cdot) : \mathcal{F} \times \mathcal{F} \rightarrow \mathbb{C}$ through the polarization identity (3.3), for which we then verify the properties of an inner product for $(\cdot, \cdot)$ in (3.2), by analogy. ■

Given the above characterization of Euclidean norms by the parallelogram identity (3.1) and the polarization identities (3.2), (3.3), approximation in Euclidean spaces is of particular importance. From the equivalence relation in Theorems 3.9 and 3.10, we can immediately draw the following conclusion.

Corollary 3.11. *Every inner product is continuous.*

Proof. Let $\mathcal{F}$ be a Euclidean space over $\mathbb{R}$ with inner product $(\cdot, \cdot)$. Moreover, let $(v_n)_{n \in \mathbb{N}} \subset \mathcal{F}$ and $(w_n)_{n \in \mathbb{N}} \subset \mathcal{F}$ be convergent sequences in $\mathcal{F}$ with limit elements $v \in \mathcal{F}$ and $w \in \mathcal{F}$. From the polarization identity (3.2) and by the continuity of the norm $\|\cdot\| = (\cdot, \cdot)^{1/2}$, from Theorem 3.4, we have

$$\begin{aligned} (v_n, w_m) &= \frac{1}{4} (\|v_n + w_m\|^2 - \|v_n - w_m\|^2) \\ &\longrightarrow \frac{1}{4} (\|v + w\|^2 - \|v - w\|^2) = (v, w) \quad \text{for } n, m \rightarrow \infty. \end{aligned}$$

For the case where $\mathcal{F}$ is a Euclidean space over $\mathbb{C}$, we show the continuity of $(\cdot, \cdot)$ from the polarization identity (3.3), by analogy. ■

Now we return to the question for the existence of best approximations. In Euclidean spaces $\mathcal{F}$ we can rely on the parallelogram identity (3.1). Moreover, we need the *completeness* of $\mathcal{F}$. On this occasion, we recall the following definition.

Definition 3.12. A complete Euclidean space is called **Hilbert³ space**. $\bigcirc$

Moreover, let us recall the notion of strictly convex sets.

Definition 3.13. A non-empty subset $\mathcal{K} \subset \mathcal{F}$ is called **convex**, if for any $u, v \in \mathcal{K}$ the straight line

$$[u, v] = \{\lambda u + (1 - \lambda)v \mid \lambda \in [0, 1]\}$$

between u and v lies in $\mathcal{K}$, i.e., if $[u, v] \subset \mathcal{K}$ for all $u, v \in \mathcal{K}$.

If for any $u, v \in \mathcal{K}$, $u \neq v$, the open straight line

$$(u, v) = \{\lambda u + (1 - \lambda)v \mid \lambda \in (0, 1)\}$$

is contained in the interior of $\mathcal{K}$, then $\mathcal{K}$ is called **strictly convex**. $\bigcirc$

Now we prove an important result concerning the existence of best approximations in Hilbert spaces.

Theorem 3.14. Let $\mathcal{F}$ be a Hilbert space with inner product $(\cdot, \cdot)$ and norm $\|\cdot\| = (\cdot, \cdot)^{1/2}$. Moreover, let $\mathcal{S} \subset \mathcal{F}$ be a closed and convex subset of $\mathcal{F}$. Then there exists for any $f \in \mathcal{F}$ a best approximation $s^* \in \mathcal{S}$ to f .

Proof. Let $(s_n)_{n \in \mathbb{N}} \subset \mathcal{S}$ be a **minimal sequence** in $\mathcal{S}$, i.e.,

$$\|s_n - f\| \longrightarrow \eta(f, \mathcal{S}) \quad \text{for } n \rightarrow \infty$$

with minimal distance $\eta \equiv \eta(f, \mathcal{S}) = \inf_{s \in \mathcal{S}} \|s - f\|$.

From the parallelogram identity (3.1) we obtain the estimate

$$\begin{aligned} \|s_n - s_m\|^2 &= 2\|s_n - f\|^2 + 2\|s_m - f\|^2 - 4\left\|\frac{s_n + s_m}{2} - f\right\|^2 \\ &\leq 2\|s_n - f\|^2 + 2\|s_m - f\|^2 - 4\eta^2. \end{aligned}$$

Therefore, for any $\varepsilon > 0$ there is one $N \equiv N(\varepsilon) \in \mathbb{N}$ satisfying

$$\|s_n - s_m\| < \varepsilon \quad \text{for all } n, m \geq N,$$

i.e., $(s_n)_{n \in \mathbb{N}}$ is a **Cauchy⁴ sequence** in the Hilbert space $\mathcal{F}$, and therefore convergent in $\mathcal{F}$. Since $\mathcal{S}$ is a closed set, the limit element s^* lies in $\mathcal{S}$, and we have

$$\eta = \lim_{n \rightarrow \infty} \|s_n - f\| = \|s^* - f\|,$$

i.e., $s^* \in \mathcal{S}$ is a best approximation to f . $\blacksquare$

³ DAVID HILBERT (1862-1943), German mathematician

⁴ AUGUSTIN-LOUIS CAUCHY (1789-1857), French mathematician

Remark 3.15. The required convexity for $\mathcal{S}$ is necessary for the result of Theorem 3.14. In order to see this, we regard the sequence space

$$\ell^2 \equiv \ell^2(\mathbb{R}) = \left\{ x = (x_k)_{k \in \mathbb{N}} \subset \mathbb{R} \left| \sum_{k=1}^{\infty} |x_k|^2 < \infty \right. \right\} \quad (3.10)$$

consisting of all square summable sequences of real numbers. The sequence space ℓ^2 , being equipped with the inner product

$$(x, y) = \sum_{k=1}^{\infty} x_k y_k \quad \text{for } x = (x_k)_{k \in \mathbb{N}}, y = (y_k)_{k \in \mathbb{N}} \in \ell^2$$

is a Hilbert space with the ℓ^2 -norm

$$\|x\|_2 := \sqrt{\sum_{k=1}^{\infty} |x_k|^2} \quad \text{for } x = (x_k)_{k \in \mathbb{N}} \in \ell^2.$$

Now we regard the subset

$$\mathcal{S} = \left\{ x^{(k)} = \left(1 + \frac{1}{k} \right) e_k \mid k \in \mathbb{N} \right\} \subset \ell^2,$$

where $e_k \in \ell^2$ is the sequence with $(e_k)_j = \delta_{jk}$, for $j, k \in \mathbb{N}$. Note that the elements $x^{(k)} \in \mathcal{S}$ are isolated in ℓ^2 , and so $\mathcal{S}$ is closed. But $\mathcal{S}$ is *not* convex. Now we have $\eta(0, \mathcal{S}) = 1$ for the minimal distance between $0 \in \ell^2$ and $\mathcal{S}$, and, moreover,

$$\|x^{(k)} - 0\|_2 > 1 \quad \text{for all } x^{(k)} \in \mathcal{S}.$$

Hence there exists no $x^{(k)} \in \mathcal{S}$ with unit distance to the origin.

Finally, we remark that the result of Theorem 3.14 does *not* generalize to Banach spaces. To see this, a counterexample can for instance be found in [42, Section 5.2]. $\square$

3.2 Uniqueness

In the following discussion, the notion of (*strict*) convexity for point sets, functions, functionals and norms plays an important role. Recall the relevant definitions for sets (see Definition 3.13) and for functions (see Definition 3.20), as these should be familiar from analysis.

Now we note some fundamental results, where $\mathcal{F}$ denotes, throughout this section, a linear space with norm $\|\cdot\|$. We start with a relevant example for a convex set.

Theorem 3.16. *Let $\mathcal{S} \subset \mathcal{F}$ be convex and $f \in \mathcal{F}$. Then the set*

$$\mathcal{S}^* \equiv \mathcal{S}^*(f, \mathcal{S}) = \{s^* \in \mathcal{S} \mid \|s^* - f\| = \inf_{s \in \mathcal{S}} \|s - f\|\} \subset \mathcal{S}$$

of best approximations $s^ \in \mathcal{S}$ to f is convex.*

Proof. Let $s_1^*, s_2^* \in \mathcal{S}^*$ be two best approximations to $f \in \mathcal{F}$. Then, for any element

$$s_\lambda^* = \lambda s_1^* + (1 - \lambda)s_2^* \in [s_1^*, s_2^*] \subset \mathcal{S} \quad \text{for } \lambda \in [0, 1] \quad (3.11)$$

we have

$$\begin{aligned} \|s_\lambda^* - f\| &= \|(\lambda s_1^* + (1 - \lambda)s_2^*) - (\lambda + (1 - \lambda))f\| \\ &= \|\lambda(s_1^* - f) + (1 - \lambda)(s_2^* - f)\| \\ &\leq \lambda\|s_1^* - f\| + (1 - \lambda)\|s_2^* - f\| \\ &= \lambda \inf_{s \in \mathcal{S}} \|s - f\| + (1 - \lambda) \inf_{s \in \mathcal{S}} \|s - f\| \\ &= \inf_{s \in \mathcal{S}} \|s - f\|, \end{aligned}$$

i.e., $s_\lambda^* = \lambda s_1^* + (1 - \lambda)s_2^* \in [s_1^*, s_2^*]$, for $\lambda \in [0, 1]$, lies in $\mathcal{S}^*$. ■

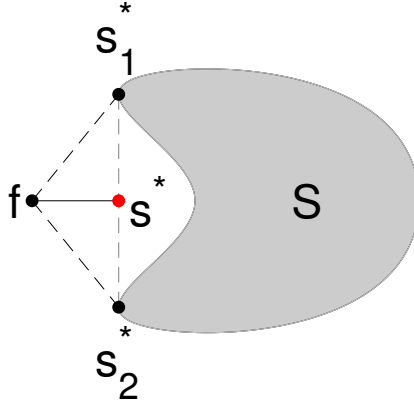


Fig. 3.3. $\mathcal{S}$ is not convex and for $s^* \notin \mathcal{S}$ we have $\|s^* - f\| < \eta(f, \mathcal{S})$, cf. Remark 3.17.

We continue with the following remarks concerning Theorem 3.16.

Remark 3.17. If $\mathcal{S} \subset \mathcal{F}$ is in the situation of Theorem 3.16 *not* convex, then any element $s^* \in [s_1^*, s_2^*]$ is at least as close to $f \in \mathcal{F}$ as s_1^* and s_2^* are, i.e.,

$$\|s^* - f\| \leq \eta \equiv \eta(f, \mathcal{S}) \quad \text{for all } s^* \in [s_1^*, s_2^*].$$

Eventually, one $s^* \in [s_1^*, s_2^*]$ could even lie closer to f than s_1^*, s_2^* , so that $\|s^* - f\| < \eta$, as the example in Figure 3.3 shows. □

Remark 3.18. For a convex subset $\mathcal{S} \subset \mathcal{F}$, and in the case of non-unique best approximations, we can explain the situation as follows. If there are at least two best approximations $s_1^* \neq s_2^*$ to f , then *all* $s^* \in [s_1^*, s_2^*]$ are contained in the set of best approximations $\mathcal{S}^*$, and so the distance between f and elements in $[s_1^*, s_2^*]$ is constant, i.e.,

$$\|s^* - f\| = \eta(f, \mathcal{S}) \quad \text{for all } s^* \in [s_1^*, s_2^*].$$

□

To further illustrate this, let us make one simple example.

Example 3.19. For $\mathcal{S} = \{x \in \mathbb{R}^2 \mid \|x\|_\infty \leq 1\}$ and $f = (2, 0)$, the set $\mathcal{S}^*$ of best approximations to f from $\mathcal{S}$ with respect to the maximum norm $\|\cdot\|_\infty$ is given by

$$\mathcal{S}^* = \{(1, \alpha) \in \mathbb{R}^2 \mid \alpha \in [-1, 1]\} \subset \mathcal{S}$$

with the minimal distance

$$\eta(f, \mathcal{S}) = \inf_{s \in \mathcal{S}} \|s - f\|_\infty = 1.$$

For $s_1^*, s_2^* \in \mathcal{S}^*$ every element $s^* \in [s_1^*, s_2^*]$ lies in $\mathcal{S}^*$ (see Figure 3.4). ◇

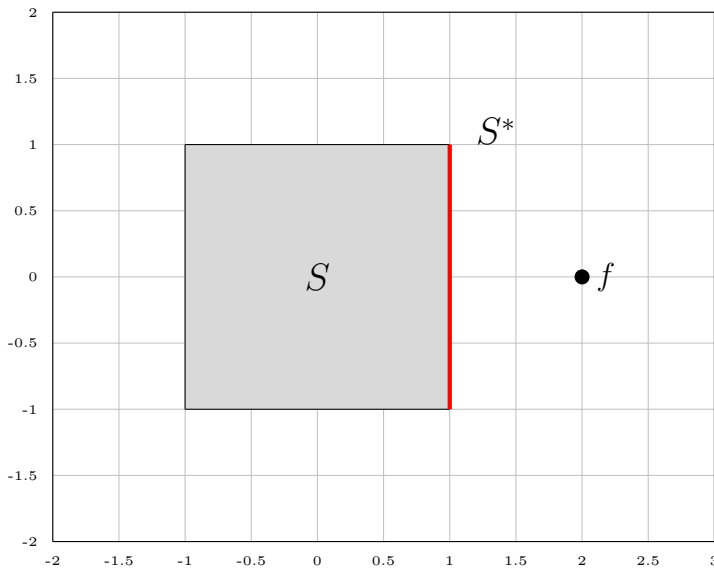


Fig. 3.4. $\mathcal{S}^* = \{(1, \alpha) \in \mathbb{R}^2 \mid \alpha \in [-1, 1]\}$ is the set of best approximations to $f = (2, 0)$ from $\mathcal{S} = \{x \in \mathbb{R}^2 \mid \|x\|_\infty \leq 1\}$ with respect to $\|\cdot\|_\infty$ (see Example 3.19).

Next, we recall the definition for (strictly) convex functions.

Definition 3.20. A function $f : [a, b] \rightarrow \mathbb{R}$ is called **convex** on an interval $[a, b] \subset \mathbb{R}$, if for all $x, y \in [a, b]$ the inequality

$$f(\lambda x + (1 - \lambda)y) \leq \lambda f(x) + (1 - \lambda)f(y) \quad \text{for all } \lambda \in [0, 1]$$

holds; f is said to be **strictly convex** on $[a, b]$, if for all $x, y \in [a, b]$, $x \neq y$, we have

$$f(\lambda x + (1 - \lambda)y) < \lambda f(x) + (1 - \lambda)f(y) \quad \text{for all } \lambda \in (0, 1).$$

○

An important property of convex functions is described by the *Jensen*⁵ inequality [39], whereby the value of a convex function, when evaluated at a finite convex combination of arguments, is bounded above by the corresponding convex combination of functions values at these arguments.

Theorem 3.21. (Jensen inequality, 1906).

Let $f : [a, b] \rightarrow \mathbb{R}$ be a convex function, and $\{x_1, \dots, x_n\} \subset [a, b]$ a set of $n \geq 2$ points. Then, the **Jensen inequality**

$$f\left(\sum_{j=1}^n \lambda_j x_j\right) \leq \sum_{j=1}^n \lambda_j f(x_j) \quad \text{for all } \lambda_j \in (0, 1) \text{ with } \sum_{j=1}^n \lambda_j = 1$$

holds. If f is strictly convex, then equality holds, if and only if all points coincide, i.e., $x_1 = \dots = x_n$.

Proof. We prove the statement of Jensen's inequality by induction on n .

Initial step: For $n = 2$, the statement of Jensen's inequality is obviously true.

Induction hypothesis: Assume the statement holds for n points $\{x_1, \dots, x_n\}$.

Induction step ($n \rightarrow n + 1$): For $n + 1$ points $\{x_1, \dots, x_n, x_{n+1}\} \subset [a, b]$ and

$$\lambda_1, \dots, \lambda_n, \lambda_{n+1} \in (0, 1) \quad \text{with } \sum_{j=1}^n \lambda_j = 1 - \lambda_{n+1}$$

we have

$$\begin{aligned} f\left(\sum_{j=1}^{n+1} \lambda_j x_j\right) &= f\left((1 - \lambda_{n+1}) \sum_{j=1}^n \frac{\lambda_j}{1 - \lambda_{n+1}} x_j + \lambda_{n+1} x_{n+1}\right) \\ &\leq (1 - \lambda_{n+1}) f\left(\sum_{j=1}^n \frac{\lambda_j}{1 - \lambda_{n+1}} x_j\right) + \lambda_{n+1} f(x_{n+1}) \end{aligned}$$

⁵ JOHAN LUDWIG JENSEN (1859-1925), Danish mathematician

by the convexity of f . By the induction hypothesis, we can further conclude

$$f\left(\sum_{j=1}^n \frac{\lambda_j}{1-\lambda_{n+1}} x_j\right) \leq \sum_{j=1}^n \frac{\lambda_j}{1-\lambda_{n+1}} f(x_j) = \frac{1}{1-\lambda_{n+1}} \sum_{j=1}^n \lambda_j f(x_j) \quad (3.12)$$

and thus, altogether,

$$f\left(\sum_{j=1}^{n+1} \lambda_j x_j\right) \leq \sum_{j=1}^{n+1} \lambda_j f(x_j). \quad (3.13)$$

If f is strictly convex, then equality holds in (3.12) only for $x_1 = \dots = x_n$ (by induction hypothesis), and, moreover, equality in (3.13) holds only for

$$x_{n+1} = \sum_{j=1}^n \frac{\lambda_j}{1-\lambda_{n+1}} x_j,$$

thus altogether only for $x_1 = \dots = x_n = x_{n+1}$. ■

Next, we introduce the convexity for functionals.

Definition 3.22. A functional $\varphi : \mathcal{F} \rightarrow \mathbb{R}$ is said to be **convex** on $\mathcal{F}$, if for all $u, v \in \mathcal{F}$ the inequality

$$\varphi(\lambda u + (1-\lambda)v) \leq \lambda \varphi(u) + (1-\lambda)\varphi(v) \quad \text{for all } \lambda \in [0, 1] \quad (3.14)$$

holds. ○

Remark 3.23. Every norm $\|\cdot\| : \mathcal{F} \rightarrow [0, \infty)$ is a convex functional on $\mathcal{F}$. Indeed, for any $u, v \in \mathcal{F}$ we find the inequality

$$\|\lambda u + (1-\lambda)v\| \leq \lambda\|u\| + (1-\lambda)\|v\| \quad \text{for all } \lambda \in [0, 1] \quad (3.15)$$

due to the triangle inequality and the homogeneity of $\|\cdot\|$. Moreover, equality in (3.15) holds for all pairs of linearly dependent elements $u, v \in \mathcal{F}$ with $u = \alpha v$ for positive scalar $\alpha > 0$, i.e., we have

$$\|\lambda \alpha v + (1-\lambda)v\| = \lambda\|\alpha v\| + (1-\lambda)\|v\| \quad \text{for all } \lambda \in [0, 1] \quad (3.16)$$

by the homogeneity of $\|\cdot\|$. □

We introduce the notion of a *strictly* convex norm classically as follows.

Definition 3.24. A norm $\|\cdot\|$ is called **strictly convex** on $\mathcal{F}$, if the unit ball $\mathcal{B} = \{u \in \mathcal{F} \mid \|u\| \leq 1\} \subset \mathcal{F}$ is *strictly convex*. ○

As we will show, not every norm is strictly convex. But before we do so, our "classical" introduction for strictly convex norms in Definition 3.24 deserves a comment.

Remark 3.25. Had we introduced the strict convexity of $\varphi : \mathcal{F} \rightarrow \mathbb{R}$ in Definition 3.22 in a straightforward manner through the inequality

$$\varphi(\lambda u + (1 - \lambda)v) < \lambda\varphi(u) + (1 - \lambda)\varphi(v) \quad \text{for all } \lambda \in (0, 1), \quad (3.17)$$

then *no* norm would be strictly convex in this particular sense! This important observation is verified by the counterexample in (3.16). $\square$

When working with strictly convex norms $\|\cdot\|$ (according to Definition 3.24), we can exclude non-uniqueness of best approximations, if $\mathcal{S} \subset \mathcal{F}$ is convex. To explain this, we need to further analyze strictly convex norms. To this end, we first prove the following useful characterization.

Theorem 3.26. *Let $\mathcal{F}$ be a linear space with norm $\|\cdot\|$. Then the following statements are equivalent.*

- (a) *The norm $\|\cdot\|$ is strictly convex.*
- (b) *The unit ball $\mathcal{B} = \{u \in \mathcal{F} \mid \|u\| \leq 1\} \subset \mathcal{F}$ is strictly convex.*
- (c) *The inequality $\|u + v\| < 2$ holds for all $u \neq v$, with $\|u\| = \|v\| = 1$.*
- (d) *The equality $\|u + v\| = \|u\| + \|v\|$, $v \neq 0$, implies $u = \alpha v$ for some $\alpha \geq 0$.*

Proof. Note that the equivalence (a) $\Leftrightarrow$ (b) holds by Definition 3.24.

(b) $\Rightarrow$ (c): The strict convexity of $\mathcal{B}$ implies $\|(u + v)/2\| < 1$ for $u \neq v$ with $\|u\| = \|v\| = 1$, and so in this case we have $\|u + v\| < 2$.

(c) $\Rightarrow$ (d): For $u = 0$ statement (d) holds with $\alpha = 0$. Now suppose $u, v \in \mathcal{F} \setminus \{0\}$ satisfy $\|u + v\| = \|u\| + \|v\|$. Without loss of generality, we may assume $\|u\| \leq \|v\|$ (otherwise we swap u and v). In this case, in the sequence of inequalities

$$\begin{aligned} 2 &\geq \left\| \frac{u}{\|u\|} + \frac{v}{\|v\|} \right\| = \left\| \left(\frac{u}{\|u\|} + \frac{v}{\|u\|} \right) - \left(\frac{v}{\|u\|} - \frac{v}{\|v\|} \right) \right\| \\ &\geq \left\| \frac{u}{\|u\|} + \frac{v}{\|u\|} \right\| - \left\| \frac{v}{\|u\|} - \frac{v}{\|v\|} \right\| = \frac{\|u + v\|}{\|u\|} - \left| \frac{1}{\|u\|} - \frac{1}{\|v\|} \right| \|v\| \\ &= \frac{\|u\| + \|v\|}{\|u\|} - \left(\frac{1}{\|u\|} - \frac{1}{\|v\|} \right) \|v\| = 2 \end{aligned}$$

equality holds everywhere, in particular

$$\left\| \frac{u}{\|u\|} + \frac{v}{\|v\|} \right\| = 2.$$

From (c) we can conclude $u/\|u\| = v/\|v\|$ and therefore

$$u = \alpha v \quad \text{for } \alpha = \frac{\|u\|}{\|v\|} > 0.$$

(d) $\Rightarrow$ (b): Suppose $u, v \in \mathcal{B}$, $u \neq v$, i.e., $\|u\| \leq 1$ and $\|v\| \leq 1$. Then we find for any $\lambda \in (0, 1)$ the inequality

$$\|\lambda u + (1 - \lambda)v\| \leq \lambda\|u\| + (1 - \lambda)\|v\| < 1,$$

provided that $\|u\| < 1$ or $\|v\| < 1$. Otherwise, i.e., if $\|u\| = \|v\| = 1$, we have

$$\|\lambda u\| + \|(1 - \lambda)v\| = \lambda\|u\| + (1 - \lambda)\|v\| = 1 \quad \text{for } \lambda \in (0, 1).$$

If $\|\lambda u + (1 - \lambda)v\| = 1$, then we have $\lambda u = \alpha(1 - \lambda)v$ for one $\alpha > 0$ from (d). Therefore, we have $u = v$, since $\|u\| = \|v\|$. This, however, is in contradiction to the assumption $u \neq v$. Therefore, we have, also for this case,

$$\|\lambda u + (1 - \lambda)v\| < 1 \quad \text{for all } \lambda \in (0, 1).$$

■

Next, we make explicit examples of strictly convex norms. A first simple example is the absolute value $|\cdot|$, taken as a norm on $\mathbb{R}$.

Remark 3.27. The absolute value $|\cdot|$ is a strictly convex norm on $\mathbb{R}$. Indeed, in the equivalence (c) of Theorem 3.26 we can only use the two points $u = -1$ and $v = 1$, where we have $|u + v| = 0 < 2$. But note that the absolute value, when regarded as a function $|\cdot| : \mathbb{R} \rightarrow \mathbb{R}$ is *not* strictly convex on $\mathbb{R}$. □

Further examples are Euclidean norms.

Theorem 3.28. *Every Euclidean norm is strictly convex.*

Proof. Let $\mathcal{F}$ be a linear space with Euclidean norm $\|\cdot\| = (\cdot, \cdot)^{1/2}$. By Theorem 3.9, the parallelogram inequality (3.1) holds in $\mathcal{F}$, and so

$$\left\| \frac{u+v}{2} \right\|^2 + \left\| \frac{u-v}{2} \right\|^2 = \frac{\|u\|^2}{2} + \frac{\|v\|^2}{2} \quad \text{for all } u, v \in \mathcal{F}.$$

For $u, v \in \mathcal{F}$, $u \neq v$, with $\|u\| = \|v\|$ we thus have

$$\left\| \frac{u+v}{2} \right\|^2 < \|u\|^2 = \|v\|^2,$$

or, $\|u+v\| < 2$ for $\|u\| = \|v\| = 1$.

By statement (c) in Theorem 3.26, we see that $\|\cdot\|$ is strictly convex. ■

Next, we regard the linear space of all bounded sequences,

$$\ell^\infty \equiv \ell^\infty(\mathbb{R}) = \left\{ x = (x_k)_{k \in \mathbb{N}} \subset \mathbb{R} \left| \sup_{k \in \mathbb{N}} |x_k| < \infty \right. \right\},$$

equipped with the ℓ^∞ -norm

$$\|x\|_\infty := \sup_{k \in \mathbb{N}} |x_k| \quad \text{for } x = (x_k)_{k \in \mathbb{N}} \in \ell^\infty.$$

Moreover, we regard for $1 \leq p < \infty$ the linear subspaces

$$\ell^p \equiv \ell^p(\mathbb{R}) = \left\{ x = (x_k)_{k \in \mathbb{N}} \subset \mathbb{R} \left| \sum_{k=1}^{\infty} |x_k|^p < \infty \right. \right\} \subset \ell^\infty, \quad (3.18)$$

equipped with the ℓ^p -norm

$$\|x\|_p := \left(\sum_{k=1}^{\infty} |x_k|^p \right)^{1/p} \quad \text{for } x = (x_k)_{k \in \mathbb{N}} \in \ell^p.$$

To further analyze the ℓ^p -norms we prove the *Hölder⁶ inequality*.

Theorem 3.29. (Hölder inequality, 1889).

Let $1 < p, q < \infty$ satisfy $1/p + 1/q = 1$. Then, the Hölder inequality

$$\|xy\|_1 \leq \|x\|_p \|y\|_q \quad \text{for all } x \in \ell^p, y \in \ell^q, \quad (3.19)$$

holds with equality in (3.19), if and only if either $x = 0$ or $y = 0$ or

$$|x_k|^{p-1} = \alpha |y_k| \quad \text{with } \alpha = \frac{\|x\|_p^{p-1}}{\|y\|_q} > 0 \quad \text{for } y \neq 0. \quad (3.20)$$

Proof. For $1 < p, q < \infty$ with $1/p + 1/q = 1$ let

$$x = (x_k)_{k \in \mathbb{N}} \in \ell^p \quad \text{and} \quad y = (y_k)_{k \in \mathbb{N}} \in \ell^q.$$

For $x = 0$ or $y = 0$ the Hölder inequality (3.19) is trivial. Now suppose $x, y \neq 0$. Then, we find for $k \in \mathbb{N}$ the estimate

$$-\log \left(\frac{1}{p} \frac{|x_k|^p}{\|x\|_p^p} + \frac{1}{q} \frac{|y_k|^q}{\|y\|_q^q} \right) \leq -\frac{1}{p} \log \left(\frac{|x_k|^p}{\|x\|_p^p} \right) - \frac{1}{q} \log \left(\frac{|y_k|^q}{\|y\|_q^q} \right) \quad (3.21)$$

by the Jensen inequality, Theorem 3.21, here applied to the strictly convex function $-\log : (0, \infty) \rightarrow \mathbb{R}$. This yields the *Young⁷ inequality*

$$\frac{|x_k y_k|}{\|x\|_p \|y\|_q} = \left(\frac{|x_k|^p}{\|x\|_p^p} \right)^{1/p} \left(\frac{|y_k|^q}{\|y\|_q^q} \right)^{1/q} \leq \frac{1}{p} \frac{|x_k|^p}{\|x\|_p^p} + \frac{1}{q} \frac{|y_k|^q}{\|y\|_q^q}. \quad (3.22)$$

Moreover, by Theorem 3.21, we have equality in (3.21), and therefore equality in (3.22), if and only if

$$\frac{|x_k|^p}{\|x\|_p^p} = \frac{|y_k|^q}{\|y\|_q^q}. \quad (3.23)$$

By $q = p/(p-1)$ we see that (3.23) is equivalent to

⁶ HÖLDER, OTTO (1859-1937), German mathematician

⁷ WILLIAM HENRY YOUNG (1863-1942), English mathematician

$$\frac{|x_k|}{\|x\|_p} = \left(\frac{|y_k|}{\|y\|_q} \right)^{1/(p-1)}. \quad (3.24)$$

Therefore, we have equality in (3.22), if and only if (3.20) holds. Summing up both sides in the Young inequality (3.22) over k , we find

$$\sum_{k=1}^{\infty} \frac{|x_k y_k|}{\|x\|_p \|y\|_q} \leq \sum_{k=1}^{\infty} \frac{1}{p} \frac{|x_k|^p}{\|x\|_p^p} + \sum_{k=1}^{\infty} \frac{1}{q} \frac{|y_k|^q}{\|y\|_q^q} = \frac{1}{p} + \frac{1}{q} = 1,$$

and this already proves the Hölder inequality (3.19), with equality, if and only if (3.20) holds for all $k \in \mathbb{N}$. $\blacksquare$

Now we can show the strict convexity of the ℓ^p -norms, for $1 < p < \infty$.

Theorem 3.30. *For $1 < p < \infty$, the ℓ^p -norm $\|\cdot\|_p$ on ℓ^p is strictly convex.*

Proof. For $1 < p < \infty$, let $1 < q < \infty$ be the conjugate Hölder exponent of p satisfying $1/p + 1/q = 1$.

For

$$x = (x_k)_{k \in \mathbb{N}} \quad \text{and} \quad y = (y_k)_{k \in \mathbb{N}} \in \ell^p,$$

where $x \neq y$ and $\|x\|_p = \|y\|_p = 1$, we wish to prove the inequality

$$\|x + y\|_p < 2, \quad (3.25)$$

in which case the norm $\|\cdot\|_p$ would be strictly convex by the equivalence statement (c) in Theorem 3.26.

For $s_k := |x_k + y_k|^{p-1}$ and $s := (s_k)_{k \in \mathbb{N}} \in \ell^q$ we have

$$\begin{aligned} \|x + y\|_p^p &= \sum_{k=1}^{\infty} |x_k + y_k| s_k \\ &\leq \sum_{k=1}^{\infty} (|x_k| s_k + |y_k| s_k) \end{aligned} \quad (3.26)$$

$$\leq \|x\|_p \|s\|_q + \|y\|_p \|s\|_q, \quad (3.27)$$

where we applied the Hölder inequality (3.19) in (3.27) twice.

By $p = (p-1)q$, we have

$$\|s\|_q = \left(\sum_{k=1}^{\infty} |x_k + y_k|^{(p-1)q} \right)^{1/q} = \left(\sum_{k=1}^{\infty} |x_k + y_k|^p \right)^{\frac{1}{p} \cdot \frac{p}{q}} = \|x + y\|_p^{p-1}$$

and this implies, in combination with (3.27), the *Minkowski*⁸ inequality

⁸ HERMANN MINKOWSKI (1864-1909), German mathematician and physicist

$$\|x + y\|_p \leq \|x\|_p + \|y\|_p \quad \text{for all } x, y \in \ell^p,$$

in particular,

$$\|x + y\|_p \leq 2 \quad \text{for } \|x\|_p = \|y\|_p = 1.$$

If $\|x + y\|_p = 2$ for $\|x\|_p = \|y\|_p = 1$, then we have equality in both (3.26) and (3.27). But equality in (3.27) is by (3.20) equivalent to the two conditions

$$|x_k|^{p-1} = \alpha |s_k| \quad \text{and} \quad |y_k|^{p-1} = \alpha |s_k| \quad \text{with } \alpha = \frac{1}{\|s\|_q},$$

which implies

$$|x_k| = |y_k| \quad \text{for all } k \in \mathbb{N}.$$

In this case, we have equality in (3.26), if and only if $\text{sgn}(x_k) = \text{sgn}(y_k)$, for all $k \in \mathbb{N}$, i.e., equality in (3.26) and (3.27) implies $x = y$.

Therefore, the inequality (3.25) holds for all $x \neq y$ with $\|x\|_p = \|y\|_p = 1$. ■

Remark 3.31. Theorem 3.30 can be generalized to L^p -norms,

$$\|u\|_p := \left(\int_{\mathbb{R}^d} |u(x)|^p dx \right)^{1/p} \quad \text{for } u \in L^p,$$

for $1 < p < \infty$, where $L^p \equiv L^p(\mathbb{R}^d)$ is the linear space of all functions whose p -th power is *Lebesgue*⁹ *integrable*. Indeed, in this case (in analogy to Theorem 3.29) the *Hölder inequality*

$$\|uv\|_1 \leq \|u\|_p \|v\|_q \quad \text{for all } u \in L^p, v \in L^q$$

holds for $1 < p, q < \infty$ satisfying $1/p + 1/q = 1$. This implies (as in the proof of Theorem 3.30) the *Minkowski inequality*

$$\|u + v\|_p \leq \|u\|_p + \|v\|_p \quad \text{for all } u, v \in L^p,$$

where for $1 < p < \infty$ we have equality, if and only if $u = \alpha v$ for some $\alpha \geq 0$ (see [35, Theorem 12.6]). Therefore, the L^p -norm $\|\cdot\|_p$, for $1 < p < \infty$, is by equivalence statement (d) in Theorem 3.26 strictly convex. □

We can conclude our statement from Remark 3.31 as follows.

Theorem 3.32. *For $1 < p < \infty$, the L^p -norm $\|\cdot\|_p$ on L^p is strictly convex.* □

But there are norms that are *not* strictly convex. Here are two examples.

⁹ HENRI LÉON LEBESGUE (1875-1941), French mathematician

Example 3.33. The ℓ^1 -norm $\|\cdot\|_1$ on ℓ^1 in (3.18), defined as

$$\|x\|_1 = \sum_{k=1}^{\infty} |x_k| \quad \text{for } x = (x_k)_{k \in \mathbb{N}} \in \ell^1,$$

is not strictly convex, since the unit ball $\mathcal{B}_1 = \{x \in \ell^1 \mid \|x\|_1 \leq 1\} \subset \ell^1$ is not strictly convex. Indeed, for any pair of two unit vectors $e_j, e_k \in \ell^1$, $j \neq k$, we have $\|e_j\|_1 = \|e_k\|_1 = 1$ and

$$\|\lambda e_j + (1 - \lambda)e_k\|_1 = \lambda + (1 - \lambda) = 1 \quad \text{for all } \lambda \in (0, 1).$$

Thus, by Theorem 3.26, statement (b), the ℓ^1 -norm $\|\cdot\|_1$ is not strictly convex.

Likewise, we show that for the linear space ℓ^∞ of all bounded sequences the ℓ^∞ -norm $\|\cdot\|_\infty$, defined as

$$\|x\|_\infty = \sup_{k \in \mathbb{N}} |x_k| \quad \text{for } x = (x_k)_{k \in \mathbb{N}} \in \ell^\infty,$$

is not strictly convex. This is because for any $e_k \in \ell^\infty$, $k \in \mathbb{N}$, and the constant sequence $\mathbf{1} = (1)_{k \in \mathbb{N}} \in \ell^\infty$, we have $\|e_k\|_\infty = \|\mathbf{1}\|_\infty = 1$ and

$$\|\lambda e_k + (1 - \lambda)\mathbf{1}\|_\infty = 1 \quad \text{for all } \lambda \in (0, 1).$$

◇

Example 3.34. For the linear space $\mathcal{C}([0, 1]^d)$ of all continuous functions on the unit cube $[0, 1]^d \subset \mathbb{R}^d$, the maximum norm $\|\cdot\|_\infty$, defined as

$$\|u\|_\infty = \max_{x \in [0, 1]^d} |u(x)| \quad \text{for } u \in \mathcal{C}([0, 1]^d),$$

is not strictly convex. To see this we take a continuous function $u_1 \in \mathcal{C}([0, 1]^d)$ satisfying $\|u_1\|_\infty = 1$ and another continuous function $u_2 \in \mathcal{C}([0, 1]^d)$ satisfying $\|u_2\|_\infty = 1$, so that $|u_1|$ and $|u_2|$ attain their maximum on $[0, 1]^d$ at one point $x^* \in [0, 1]^d$, respectively, i.e.,

$$\|u_1\|_\infty = \max_{x \in [0, 1]^d} |u_1(x)| = |u_1(x^*)| = |u_2(x^*)| = \max_{x \in [0, 1]^d} |u_2(x)| = \|u_2\|_\infty = 1.$$

This then implies for $u_\lambda = \lambda u_1 + (1 - \lambda)u_2 \in (u_1, u_2)$, with $\lambda \in (0, 1)$,

$$|u_\lambda(x)| \leq \lambda |u_1(x)| + (1 - \lambda) |u_2(x)| \leq 1 \quad \text{for all } x \in [0, 1]^d$$

where equality holds for $x = x^*$, whereby $\|u_\lambda\|_\infty = 1$ for all $\lambda \in (0, 1)$.

In this case, the unit ball $\mathcal{B} = \{u \in \mathcal{C}([0, 1]^d) \mid \|u\|_\infty \leq 1\}$ is not strictly convex, i.e., $\|\cdot\|_\infty$ is not strictly convex by statement (b) in Theorem 3.26.

To make an explicit example for the required functions u_1 and u_2 , we take the *geometric mean* $u_g \in \mathcal{C}([0, 1]^d)$ and the *arithmetic mean* $u_a \in \mathcal{C}([0, 1]^d)$,

$$u_g(x) = \sqrt[d]{x_1 \cdot \dots \cdot x_d} \leq \frac{x_1 + \dots + x_d}{d} = u_a(x),$$

for $x = (x_1, \dots, x_d) \in [0, 1]^d$. Obviously, we have $\|u_g\|_\infty = \|u_a\|_\infty = 1$, where u_g and u_a attain their unique maximum on $[0, 1]^d$ at $\mathbf{1} = (1, \dots, 1) \in [0, 1]^d$. $\diamond$

Now we consider the Euclidean space $\mathbb{R}^d$, for $d \in \mathbb{N}$, as a linear subspace of the sequence space ℓ^p , $1 \leq p \leq \infty$, via the canonical embedding $i: \mathbb{R}^d \hookrightarrow \ell^p$,

$$x = (x_1, \dots, x_d)^T \in \mathbb{R}^d \longmapsto i(x) = (x_1, \dots, x_d, 0, \dots, 0, \dots) \in \ell^p.$$

This allows us to formulate the following statements concerning the strict convexity of the ℓ^p -norms $\|\cdot\|_p$ on $\mathbb{R}^d$.

Corollary 3.35. *For the ℓ^p -norms $\|\cdot\|_p$ on $\mathbb{R}^d$, defined as*

$$\|x\|_p^p = \sum_{k=1}^d |x_k|^p \text{ for } 1 \leq p < \infty \quad \text{and} \quad \|x\|_\infty = \max_{1 \leq k \leq d} |x_k|$$

the following statements are true.

- (a) *For $1 < p < \infty$, the ℓ^p -norms $\|\cdot\|_p$ are strictly convex on $\mathbb{R}^d$.*
- (b) *For $d > 1$, the ℓ^1 -norm $\|\cdot\|_1$ is not strictly convex on $\mathbb{R}^d$.*
- (c) *For $d > 1$, the ℓ^∞ -norm $\|\cdot\|_\infty$ is not strictly convex on $\mathbb{R}^d$.* ■

Remark 3.36. In statements (b), (c) of Corollary 3.35, we excluded the case $d = 1$, since in this univariate setting the norms $\|\cdot\|_1$ and $\|\cdot\|_\infty$ coincide with the strictly convex norm $|\cdot|$ on $\mathbb{R}$ (see Remark 3.27). □

Now we formulate the main result of this section.

Theorem 3.37. *Let $\mathcal{F}$ be a linear space, equipped with a strictly convex norm $\|\cdot\|$. Moreover, assume $\mathcal{S} \subset \mathcal{F}$ is convex and $f \in \mathcal{F}$. If there exists a best approximation $s^* \in \mathcal{S}$ to f , then s^* is unique.*

Proof. Suppose $s_1^*, s_2^* \in \mathcal{S}$ are two different best approximations to f from $\mathcal{S}$, i.e., $s_1^* \neq s_2^*$. Then we have

$$\|s_1^* - f\| = \|s_2^* - f\| = \inf_{s \in \mathcal{S}} \|s - f\|,$$

which, in combination with the strict convexity of the norm $\|\cdot\|$, implies

$$\left\| \frac{s_1^* + s_2^*}{2} - f \right\| = \left\| \frac{(s_1^* - f) + (s_2^* - f)}{2} \right\| < \|s_1^* - f\| = \|s_2^* - f\|. \quad (3.28)$$

Due to the assumed convexity for $\mathcal{S}$, the element $s^* = (s_1^* + s_2^*)/2$ lies in $\mathcal{S}$. Moreover, s^* is closer to f than s_1^* and s_2^* , by (3.28). But this is in contradiction to the optimality of s_1^* and s_2^* . ■

We remark that the strict convexity of the norm $\|\cdot\|$ gives, in combination with the convexity of $\mathcal{S} \subset \mathcal{F}$, only a *sufficient* condition for the uniqueness of the best approximation. Now we show that this condition is not necessary. To this end, we make a simple example.

Example 3.38. We regard the maximum norm $\|\cdot\|_\infty$ on $\mathcal{F} = \mathbb{R}^2$. Moreover, we let $f = (0, 1) \in \mathbb{R}^2$ and $\mathcal{S} = \{(\alpha, \alpha) \mid \alpha \in \mathbb{R}\} \subset \mathbb{R}^2$. Then, $s^* = (\frac{1}{2}, \frac{1}{2}) \in \mathcal{S}$ is the *unique* best approximation to f from $\mathcal{S}$ with respect to $\|\cdot\|_\infty$, although $\|\cdot\|_\infty$ is *not* strictly convex (see Figure 3.5). $\diamond$



Fig. 3.5. $s^* = (\frac{1}{2}, \frac{1}{2}) \in \mathcal{S} = \{(\alpha, \alpha) \mid \alpha \in \mathbb{R}\}$ is the *unique* best approximation to $f = (0, 1)$ w.r.t. $\|\cdot\|_\infty$, although $\|\cdot\|_\infty$ is *not* strictly convex (see Example 3.38).

We finally summarize our discussion concerning the uniqueness of best approximations, where we note three immediate conclusions from Theorem 3.37.

Corollary 3.39. Let $\mathcal{F}$ be a Euclidean space and $\mathcal{S} \subset \mathcal{F}$ be convex. Then there is for any $f \in \mathcal{F}$ at most one best approximation $s^* \in \mathcal{S}$ to f . $\blacksquare$

Corollary 3.40. Let $\mathcal{S} \subset L^p$ be convex for $1 < p < \infty$. Then there is for any $f \in L^p$ at most one best approximation $s^* \in \mathcal{S}$ to f w.r.t. $\|\cdot\|_p$. $\blacksquare$

Corollary 3.41. Let $\mathcal{S} \subset \ell^p$ be convex for $1 < p < \infty$. Then there is for any $f \in \ell^p$ at most one best approximation $s^* \in \mathcal{S}$ to f w.r.t. $\|\cdot\|_p$. $\blacksquare$

We finally formulate an important result concerning the approximation of continuous functions from $\mathcal{C}[-1, 1]$ by approximation spaces $\mathcal{S} \subset \mathcal{C}[-1, 1]$ that are *invariant under reflections of the argument* (in short: *reflection-invariant*), i.e., for any $s(x) \in \mathcal{S}$ the function $s(-x)$ lies also in $\mathcal{S}$. For example, the linear space $\mathcal{P}_n$ of all algebraic polynomials of degree at most $n \in \mathbb{N}_0$ is reflection-invariant. For the following observation, the uniqueness of the best approximation plays an important role.

Proposition 3.42. *Let $f \in \mathcal{C}[-1, 1]$ be an even function and, moreover, let $\mathcal{S} \subset \mathcal{C}[-1, 1]$ be a reflection-invariant subset of $\mathcal{C}[-1, 1]$. If there exists a unique best approximation $s_p^* \in \mathcal{S}$ to f with respect to the L^p -norm $\|\cdot\|_p$, for $1 \leq p \leq \infty$, then s_p^* to f is an even function.*

Proof. Let $f \in \mathcal{C}[-1, 1]$ be an even function, i.e., $f(x) = f(-x)$ for all $x \in [-1, 1]$. Moreover, let $s_p^* \in \mathcal{S}$ be the unique best approximation to f with respect to $\|\cdot\|_p$, for $1 \leq p \leq \infty$. We regard the *reflected* function r_p^* for s_p^* , defined as

$$r_p^*(x) = s_p^*(-x) \quad \text{for } x \in [-1, 1].$$

By our assumption we have r_p^* in $\mathcal{S}$.

Case $p = \infty$: For the distance between r_∞^* and f with respect to $\|\cdot\|_\infty$,

$$\begin{aligned} \|r_\infty^* - f\|_\infty &= \max_{x \in [-1, 1]} |r_\infty^*(x) - f(x)| = \max_{x \in [-1, 1]} |r_\infty^*(-x) - f(-x)| \\ &= \max_{x \in [-1, 1]} |s_\infty^*(x) - f(x)| = \|s_\infty^* - f\|_\infty = \eta_\infty(f, \mathcal{S}), \end{aligned}$$

we obtain the minimal distance between f and $\mathcal{S}$ with respect to $\|\cdot\|_\infty$, i.e., $r_\infty^* \in \mathcal{S}$ is the best approximation to f . Now the uniqueness of the best approximation implies our statement $s_\infty^*(x) = r_\infty^*(x)$, or, $s_\infty^*(x) = s_\infty^*(-x)$ for all $x \in [-1, 1]$, i.e., s_∞^* is an even function on $[-1, 1]$.

Case $1 \leq p < \infty$: In this case, we regard the distance between r_p^* and f in the L^p -Norm,

$$\begin{aligned} \|r_p^* - f\|_p^p &= \int_{-1}^1 |r_p^*(x) - f(x)|^p dx = \int_{-1}^1 |r_p^*(-x) - f(-x)|^p dx \\ &= \int_{-1}^1 |s_p^*(x) - f(x)|^p dx = \|s_p^* - f\|_p^p = \eta_p^p(f, \mathcal{S}), \end{aligned}$$

whereby we get the minimal distance $\eta_p(f, \mathcal{S})$ between f and $\mathcal{S}$ with respect to $\|\cdot\|_p$. Again, by the uniqueness of the best approximation we obtain the stated result by

$$s_p^*(x) = r_p^*(x) = s_p^*(-x) \quad \text{for all } x \in [-1, 1].$$

■

For an alternative proof of Proposition 3.42, we refer to Exercise 3.74.

3.3 Dual Characterization

In this section and in the following section, we develop necessary and sufficient conditions to characterize best approximations. We begin with *dual* characterizations. To this end, let $\mathcal{F}$ be a normed linear space. We introduce the **topological dual**, or, the **dual space** of $\mathcal{F}$ as usual by

$$\mathcal{F}' = \{ \varphi : \mathcal{F} \longrightarrow \mathbb{R} \mid \varphi \text{ linear and continuous} \}.$$

The elements from the linear space $\mathcal{F}'$ are called **dual functionals**. On this occasion, we recall the notions of *linearity*, *continuity* and *boundedness* of functionals. We start with linearity.

Definition 3.43. A functional $\varphi : \mathcal{F} \longrightarrow \mathbb{R}$ is called **linear** on $\mathcal{F}$, if

$$\varphi(\alpha u + \beta v) = \alpha \varphi(u) + \beta \varphi(v) \quad \text{for all } u, v \in \mathcal{F} \text{ and all } \alpha, \beta \in \mathbb{R}.$$

○

We had introduced continuity already in Definition 3.3. Next, we turn to the boundedness of functionals. In the following discussion, $\mathcal{F}$ denotes a linear space with norm $\| \cdot \|$.

Definition 3.44. A functional $\varphi : \mathcal{F} \longrightarrow \mathbb{R}$ is said to be **bounded** on $\mathcal{F}$, if there exists a constant $C \equiv C_\varphi > 0$ satisfying

$$|\varphi(u)| \leq C \|u\| \quad \text{for all } u \in \mathcal{F}. \quad (3.29)$$

We call such a constant C **upper bound** for φ .

○

Now we can introduce a norm for the dual space $\mathcal{F}'$, by using the norm $\| \cdot \|$ of $\mathcal{F}$. To this end, we take for any functional $\varphi \in \mathcal{F}'$ the *smallest* upper bound $C \equiv C_\varphi$ in (3.29). To be more precise on this, we define by

$$\|\varphi\| = \sup_{\substack{u \in \mathcal{F} \\ u \neq 0}} \frac{|\varphi(u)|}{\|u\|} = \sup_{\substack{u \in \mathcal{F} \\ \|u\|=1}} |\varphi(u)|$$

a mapping $\| \cdot \| : \mathcal{F}' \longrightarrow \mathbb{R}$. As it can be verified by elementary calculations, $\| \cdot \|$ is a norm on $\mathcal{F}'$, according to Definition 1.1. In other words, the dual space $\mathcal{F}'$ is a linear space with norm $\| \cdot \|$.

The following result for linear functionals is quite important.

Theorem 3.45. For a linear functional $\varphi : \mathcal{F} \longrightarrow \mathbb{R}$, the following statements are equivalent.

- (a) φ is continuous at one $u_0 \in \mathcal{F}$.
- (b) φ is continuous on $\mathcal{F}$.
- (c) φ is bounded on $\mathcal{F}$.

Proof. (a) $\Rightarrow$ (b): Let φ be continuous at $u_0 \in \mathcal{F}$, and, moreover, let $(u_n)_{n \in \mathbb{N}}$ be a convergent sequence in $\mathcal{F}$ with limit $u \in \mathcal{F}$. Then we have

$$\varphi(u_n) = \varphi(u_n - u + u_0) + \varphi(u - u_0) \longrightarrow \varphi(u_0) + \varphi(u - u_0) = \varphi(u) \quad \text{for } n \rightarrow \infty.$$

Therefore, φ is continuous at every $u \in \mathcal{F}$, i.e., φ is continuous on $\mathcal{F}$.

The implication (b) $\Rightarrow$ (a) is trivial, so the equivalence (a) $\Leftrightarrow$ (b) is shown.

(c) $\Rightarrow$ (b): Let φ be bounded on $\mathcal{F}$, i.e., we have (3.29) for some $C > 0$. This implies $\varphi(u_n) \longrightarrow 0$, $n \rightarrow \infty$, for every zero sequence $(u_n)_{n \in \mathbb{N}}$ in $\mathcal{F}$, and so φ is continuous at zero. By the equivalence (a) $\Leftrightarrow$ (b) is φ continuous on $\mathcal{F}$.

(b) $\Rightarrow$ (c): Let φ be continuous on $\mathcal{F}$. Suppose φ is not bounded on $\mathcal{F}$. Then, there is a sequence $(u_n)_{n \in \mathbb{N}}$ in $\mathcal{F}$ satisfying

$$\|u_n\| = 1 \quad \text{and} \quad |\varphi(u_n)| > n \quad \text{for all } n \in \mathbb{N},$$

since otherwise there would exist an upper bound $N \in \mathbb{N}$ for φ (i.e., φ would be bounded). In this case, the sequence $(v_n)_{n \in \mathbb{N}}$, defined as

$$v_n = \frac{u_n}{|\varphi(u_n)|} \quad \text{for } n \in \mathbb{N},$$

is a zero sequence in $\mathcal{F}$ by

$$\|v_n\| = \frac{1}{|\varphi(u_n)|} \longrightarrow 0 \quad \text{for } n \rightarrow \infty,$$

and so, by continuity of φ , we have

$$\varphi(v_n) \longrightarrow \varphi(0) = 0 \quad \text{for } n \rightarrow \infty.$$

But this is in contradiction to $|\varphi(v_n)| = 1$ for all $n \in \mathbb{N}$. ■

Now we are in a position where we can formulate a sufficient condition for the dual characterization of best approximations.

Theorem 3.46. *Let $\mathcal{S} \subset \mathcal{F}$ be a non-empty subset of $\mathcal{F}$. Moreover, let $f \in \mathcal{F}$ and $s^* \in \mathcal{S}$. Suppose that $\varphi \in \mathcal{F}'$ is a dual functional satisfying the following properties.*

- (a) $\|\varphi\| = 1$.
- (b) $\varphi(s^* - f) = \|s^* - f\|$.
- (c) $\varphi(s - s^*) \geq 0$ for all $s \in \mathcal{S}$.

Then s^ is a best approximation to f .*

Proof. For $s \in \mathcal{S}$, we have $\varphi(s - f) \leq \|s - f\|$, due to (a). Moreover, we have

$$\|s - f\| \geq \varphi(s - f) = \varphi(s - s^*) + \varphi(s^* - f) \geq \|s^* - f\|$$

by (b) and (c). Therefore, s^* is a best approximation to f . ■

Note that the above characterization in Theorem 3.46 only requires $\mathcal{S}$ to be non-empty. However, if we assume $\mathcal{S} \subset \mathcal{F}$ to be convex, then we can show that the sufficient condition in Theorem 3.46 is also necessary. To this end, we need the following *separation theorem for convex sets*, which can be viewed as a geometric implication from the well-known *Hahn¹⁰-Banach¹¹ theorem* (see [33, Section 16.1]) which was proven by Mazur¹² in [3].

Theorem 3.47. (Banach-Mazur separation theorem, 1933).

Let $\mathcal{K}_1, \mathcal{K}_2 \subset \mathcal{F}$ be two non-empty, disjoint and convex subsets in a normed linear space $\mathcal{F}$. Moreover, suppose $\mathcal{K}_1$ is an open set. Then there exists a separating functional $\varphi \in \mathcal{F}'$ for $\mathcal{K}_1$ and $\mathcal{K}_2$, i.e., we have

$$\varphi(u_1) < \varphi(u_2) \quad \text{for all } u_1 \in \mathcal{K}_1, u_2 \in \mathcal{K}_2.$$

□

On the Banach-Mazur separation theorem, we can formulate a sufficient and necessary condition for the dual characterization of best approximations.

Theorem 3.48. Let $\mathcal{S} \subset \mathcal{F}$ be a convex set in $\mathcal{F}$. Moreover, suppose $f \in \mathcal{F} \setminus \mathcal{S}$. Then, $s^* \in \mathcal{S}$ is a best approximation to f , if and only if there exists a dual functional $\varphi \in \mathcal{F}'$ satisfying the following properties.

- (a) $\|\varphi\| = 1$.
- (b) $\varphi(s^* - f) = \|s^* - f\|$.
- (c) $\varphi(s - s^*) \geq 0$ for all $s \in \mathcal{S}$.

Proof. Note that the sufficiency of the statement is covered by Theorem 3.46.

To prove the necessity, suppose that $s^* \in \mathcal{S}$ is a best approximation to f . Regard the open ball

$$\mathcal{B}_\eta(f) = \{u \in \mathcal{F} \mid \|u - f\| < \|s^* - f\|\} \subset \mathcal{F}$$

around f with radius $\eta = \|s^* - f\|$. Note that for $\mathcal{K}_1 = \mathcal{B}_\eta(f)$ and $\mathcal{K}_2 = \mathcal{S}$ the assumptions of the Banach-Mazur separation theorem, Theorem 3.47, are satisfied. Therefore, there is a separating functional $\varphi \in \mathcal{F}'$ with

$$\varphi(u) < \varphi(s) \quad \text{for all } u \in \mathcal{B}_\eta(f) \text{ and } s \in \mathcal{S}. \quad (3.30)$$

Now let $(u_n)_{n \in \mathbb{N}} \subset \mathcal{B}_\eta(f)$ be a convergent sequence with limit element s^* , i.e., $u_n \rightarrow s^*$ for $n \rightarrow \infty$. By the continuity of φ , this implies

$$\varphi(u_n) \rightarrow \varphi(s^*) = \inf_{s \in \mathcal{S}} \varphi(s),$$

¹⁰ HANS HAHN (1879-1934), Austrian mathematician and philosopher

¹¹ STEFAN BANACH (1892-1945), Polish mathematician

¹² STANISŁAW MAZUR (1905-1981), Polish mathematician

i.e., $\varphi(s^*) \leq \varphi(s)$ for all $s \in \mathcal{S}$, and so φ has property (c).

To show properties (a) and (b), let $v \in \mathcal{F}$ with $\|v\| < 1$. Then, $u = \eta v + f$ lies in $\mathcal{B}_\eta(f)$. With (3.30) and by the linearity of φ , we have

$$\varphi(v) = \varphi\left(\frac{u - f}{\|s^* - f\|}\right) < \varphi\left(\frac{s^* - f}{\|s^* - f\|}\right).$$

This implies

$$\|\varphi\| = \sup_{\|v\| \leq 1} |\varphi(v)| \leq \varphi\left(\frac{s^* - f}{\|s^* - f\|}\right)$$

and, moreover, by using the continuity of φ once more, we have

$$\|\varphi\| = \frac{\varphi(s^* - f)}{\|s^* - f\|} \iff \varphi(s^* - f) = \|\varphi\| \cdot \|s^* - f\|.$$

If we finally normalize the length of $\varphi \neq 0$, by scaling φ to unit norm, i.e., $\|\varphi\| = 1$, then φ satisfies properties (a) and (b). $\blacksquare$

3.4 Direct Characterization

In this section, we develop necessary and sufficient conditions for the minimization of convex functionals. We then apply these conditions to norms to obtain useful characterizations for best approximations. To this end, we work with *Gâteaux¹³ derivatives* to compute directional derivatives for relevant norms. In the following discussion, $\mathcal{F}$ denotes a linear space.

Definition 3.49. For a functional $\varphi : \mathcal{F} \rightarrow \mathbb{R}$,

$$\varphi'_+(u, v) := \lim_{h \searrow 0} \frac{1}{h} (\varphi(u + hv) - \varphi(u)) \quad \text{for } u, v \in \mathcal{F} \quad (3.31)$$

is said to be the **Gâteaux derivative** of φ at u in direction v , provided that the limit on the right hand side in (3.31) exists. $\circ$

For convex functionals $\varphi : \mathcal{F} \rightarrow \mathbb{R}$ we can show that the limit on the right hand side in (3.31) exists.

Theorem 3.50. Let $\varphi : \mathcal{F} \rightarrow \mathbb{R}$ be a convex functional. Then, the Gâteaux derivative $\varphi'_+(u, v)$ exists for all $u, v \in \mathcal{F}$. Moreover, the inequality

$$-\varphi'_+(u, -v) \leq \varphi'_+(u, v) \quad \text{for all } u, v \in \mathcal{F}$$

holds.

¹³ RENÉ GÂTEAUX (1889-1914), French mathematician